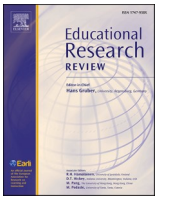
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Review



A systematic review of bilingual education teachers' competences

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ABSTRACT

Ongoing globalisation processes call forth a need to foster students' intercultural competences and language skills. Simultaneously, teachers face an increasing diversity of students' first languages and cultures within classrooms, demanding responsive and integrative practices from them which address all students. Bilingual education is one possible format in which these challenges can be met; however, it requires a high level of teacher professionalism. This systematic literature review harvests international research on bilingual education teachers published between 1995 and 2020. It compares frameworks and research on bilingual education teachers' required competences for secondary education and generates a professional competence model for bilingual education teachers. This systematic review of 79 individual reports identifies a total of 16 converging competences, many focusing on language proficiency and pedagogical/psychological knowledge. Furthermore, it yields important implications for bilingual teacher training such as the fostering of (academic) language proficiency, cooperation skills and knowledge of bilingual education research.

1. The importance of bilingual education teachers' competences: introduction and theoretical background

In the light of the growing number of diverse learners in school (OECD, 2018), bilingual education is becoming a more frequent teaching format. In principle, bilingual education describes the use of two or more languages for teaching and learning (Wright & Baker, 2017). The overall aims of bilingual education are "the assimilation of immigrants, helping children to gain employment through multilingual [and multicultural] competences, increasing school achievement [or] helping to preserve a minority language" (Wright & Baker, 2017, p. 66). In North America, immersion programs, predominantly for second language learning, and dual-language programs, mostly for minority students (Genesee & Lindholm-Leary, 2013) are widely implemented. Both are categorised as content-based instruction (CBI), which combines content and language to enable language learning through actively using the language when learning content (Genesee & Lindholm-Leary, 2013). Content and language integrated learning (CLIL) is the European implementation of bilingual education. CLIL took root around 1994 (Marsh et al., 2001) and is defined as "a dual-focused educational approach in which an additional language is used for the learning and teaching of both content *and* language" (Coyle et al., 2010, p. 1). The differences between these forms of bilingual education are (controversially) discussed, however, we follow Cenoz (2015) and Dalton-Puffer and Nikula (2014) in their conclusion that there is important commonness between CLIL, immersion or CBI in general, namely that teachers' pedagogical practices and challenges are similar (Dalton-Puffer & Nikula, 2014; Genesee & Lindholm-Leary, 2013). This also shows that there is a disconnection in research between CLIL and CBI. Accordingly, this review aims at integrating the research streams with special focus on bilingual education teachers' (BETs') competences. It can be assumed that

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(bilingual education) researchers, teachers and administrators can gain new insights and, therefore, profit from a joint assessment of these forms of programs. We provide such an assessment in this review since this assumption allows us to include a larger number of studies (see below).

Effective bilingual education programs and implementation guidelines around the world emphasise teacher qualification/training as important prerequisite for a program's success (e.g., Barrios & Milla Lara, 2020; Henderson & Palmer, 2020). This is not surprising since research on the relationship between teachers' competences and students' learning outcomes has shown that teaching quality and teacher certification are the main in-school factors responsible for student success (Darling-Hammond, 2000; Hattie, 2008; Maulana et al., 2021). Competences comprise a collection of prerequisites for successful domain-specific actions which are available on an individual or inter-individual level (Weinert, 2001). They include knowledge, skills and values "leading to effective action in situation in a particular domain" (Caena, 2014, p. 315; Deakin Crick, 2008). As a theoretical background for this paper, the generic and multidimensional model of teachers' professional competence (Baumert & Kunter, 2013) will be used. The model combines the competence definition of Weinert (2001) with Shulman's (1986; 1987) and Bromme's (2001) categories of teachers' knowledge. Additionally, it assumes that competences are not innate but can be learned over time (Kunter et al., 2013). The model includes motivation, self-regulation, beliefs/values/goals regarding learning and professional knowledge (Fig. 1). The latter comprises content knowledge, pedagogical content knowledge, pedagogical/psychological knowledge, organisational knowledge and counselling knowledge (Baumert & Kunter, 2013). This model of teachers' generic professional competence provides an appropriate theoretical background for our systematic review as it enables to reveal and emphasise specific competence requirements of BETs.

Although some CLIL-specific frameworks for teachers' competences have been derived theoretically (Bertaux et al., 2010; Marsh et al., 2011), their revision through comparison with other bilingual and generic professional competence frameworks, as well as with complementary empirical or theoretical research on the competences of BETs is lacking. Above all, existing studies on BETs' competences often only focus on one aspect of overall competence. Consequently, an overall assessment of BETs' competences is still a research gap. Against this background, we systematically reviewed the literature with the aim to summarise and discuss findings regarding BETs' professional competences. Accordingly, the research question (RQ) "How are the competences of BETs in secondary schools conceptualised within research?" will guide this systematic literature review. Based on the theoretical background model, we analyse content-based types of bilingual education from all around the world to create a comprehensive conceptualisation. Although research on immersion teaching, in particular, highlighted the crucial significance of long-term engagement starting at the elementary level for successful bilingualism/biliteracy among students (Gándara & Escamilla, 2017; McIvor & McCarty, 2017; Wright & Baker, 2017), our focus was on BETs from secondary schools, i.e., from middle to high school, for the following reasons. First, primary programs often incorporate transitional bilingual education, which transitions students early on to English-only instruction, after a short period of bilingual teaching. Transitional bilingual education has been shown to be the least successful form of bilingual education (Collier & Thomas, 2004; Herrera, 2020). Also, secondary education teachers face serious challenges dealing with English language learners, due to a lack of support and preparation (Santibañez & Gándara, 2018).

Overall, this paper extends previous research on bilingual teacher education since it is the first attempt to systematically synthesise previous research to provide a comprehensive model of BETs' competences, encouraging a reflection on required professional competences in the field. It also provides practical implications for bilingual teacher education programs. Thus, this systematic review contributes to the literature on bilingual education (e.g., Rubio-Alcalá et al., 2019) as well as to that on teacher professionalism more generally (Kennedy, 2016; Opfer & Pedder, 2011).

2. Method

This systematic review is guided by the Preferred Reporting Items for Systematic Reviews and Meta-Analyses' (PRISMA) 2020 statement (Page, McKenzie et al., 2021). Preliminary non-systematic searches were carried out in September 2020 for the researchers to identify different terminologies of this field of research and determine the search terms. The review aims, as well as the inclusion and exclusion criteria in Table 1, were, then, developed. The year 1995 was chosen as a starting point because CLIL was then coming into widespread use and tendencies arose to counter the scarceness of research in the field of bilingual students' teacher efficacy (Garcia, 1991) and minority teachers in general (Villegas et al., 1993). Furthermore, the implementation of CLIL in Europe in 1994 gave rise to a greater research interest in bilingual education. Therefore, the review's time interval ranges from 1995 to 2020.

In December 2020, advanced searches of the databases Education Source, ERIC, MLA International Bibliography (via EBSCOhost) and ProQuest Dissertations were undertaken using the search string ("bilingual education" OR CLIL OR "bilingual teaching" OR "bilingual teacher") AND ("teacher training" OR "teacher education" OR knowledge OR competenc*). Furthermore, the first 100 results from Google Scholar were included in our initial search. We chose these databases, since e.g. Google Scholar or ERIC are broader as well as more encompassing than other databases like Web of Science (Alexander, 2020), which fits well with our goal of representing and including literature as diverse as possible. Additionally, the German database Education Research Portal was searched using the search string ("bilingualer Unterricht" oder CLIL) und (Lehrerbildung oder Kompetenz oder Wissen). We did not include additional forms of education, for instance, CBI or dual-language education since we consider bilingual education to be an umbrella term for all such forms. However, we included CLIL in our search terms because some researchers diverge from our assumption and consider CLIL to be an umbrella term for all forms of bilingual education (Mehisto et al., 2008). The searches returned 4298 records after the removal of duplicates through the reference management programs Citavi, Endnote, Mendeley and a final manual sorting for records in languages other than English or German.

The extracted titles and abstracts were collaboratively screened by three independent raters, based on the predefined inclusion and exclusion criteria in Table 1 and through the systematic review web application Rayyan (Ouzzani et al., 2016). All records were

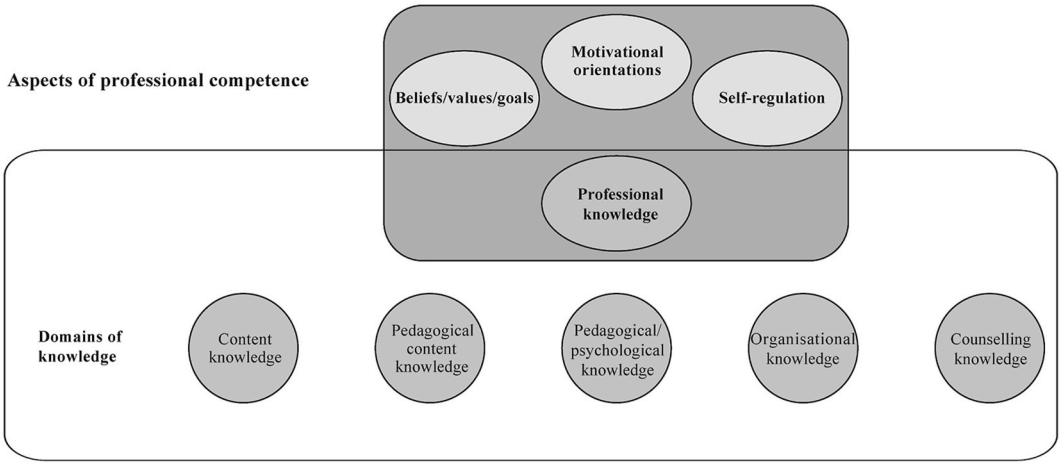


Fig. 1. Model of teachers' professional competence (adapted from Baumert & Kunter, 2013, p. 29).

Table 1
Inclusion and exclusion criteria applied to the review.

Type	Inclusion	Exclusion
Time	January 1, 1995 to December 02, 2020	Any other points in time
Target	Secondary BETs (pre-service, trainee or in-service)	Students, parents, principals, paraeducators or teaching assistants
Publication	Peer reviewed journal articles and conference proceedings, book chapters & dissertations	Presentations or policy reports
Study	Empirical studies & theoretical contributions	Autoethnographic recommendations
Content	RQ1: Bilingual teachers' competences & training needs	Reports on shortages, bilingual education in general, teaching materials or guides
Setting	Secondary schools & teacher training programs	Preschool/kindergarten, primary education, higher and adult education in general (except for teacher education)
Setting	Bilingual education forms: CLIL, CBI (immersion & dual-language programs)	Transitional bilingual education, foreign or second language education, special education
Language	English or German	Any other languages
Acquisition	Full text is available	No full text is available

screened by the first author of this paper and one of the other two raters. The reports[1] were then checked for full-text availability, and 70 non-available reports were excluded (see Fig. 2). Afterwards, the reports were read in full by two raters to further compress the selection. Detailed information on all fully read and included papers was entered into an Excel spreadsheet (e.g., publication year, authors, form of bilingual education, type of study; see also Table 2). For good measure, a backward search was conducted in which the bibliographies of included and recently published reports were reviewed and assessed so any important papers which had not yet been identified in the systematic review could be included. These reviews or theoretical contributions on competence frameworks and competence research were handled as follows: First, the references were backwards-searched and checked for inclusion. If the references were available and suitable, they were included. If, however, the referenced literature was not – or only partly – available, or in languages other than English and German, only the originally found review or theoretical contribution itself was included. The quality of the included studies, which were not peer reviewed during the publication process, was evaluated using the Critical Appraisal Skills Programme for qualitative studies (Critical Appraisal Skills Programme, 2018). The outcome of this appraisal is summarised in Table S1 (online only). Included and non-peer reviewed theoretical works were not appraised due to the lack of an appropriate appraisal framework.

In general, the raters cooperated during all steps of the selection and discussed disagreements regarding the inclusion or exclusion of papers based on their title and abstract and, later, their full text, until 100% agreement was reached on the included papers. The initial interrater agreement was 82.6 per cent for the screening of titles and abstracts and 87.9 per cent for full texts. After the final selection of the revised sample of papers, the report information in the Excel spreadsheet on competences and training programs was analysed and grouped into categories.

[1] A report is defined as "[a] document (paper or electronic) supplying information about a particular study. It could be a journal article, preprint, conference abstract, study register entry, clinical study report, dissertation, unpublished manuscript, government report, or any other document providing relevant information" (Page, Moher et al., 2021, p. 2).

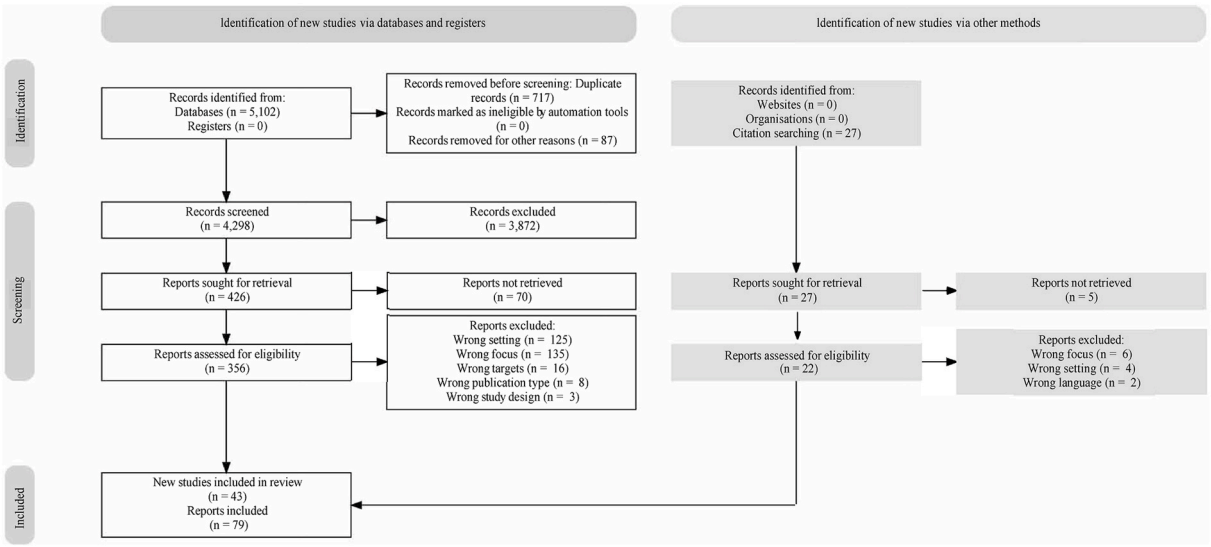


Fig. 2. Adjusted PRISMA flow diagram (Page, McKenzie, et al., 2021) created using the flow diagram tool (Haddaway et al., 2020)
Note. The included reports encompasse English (n = 76) and German reports (n = 3).

3. Results

In this section, the included reports and their results are synthesised. First, the selection process is depicted in a PRISMA flow diagram (see Fig. 2). Then, an overview of the descriptive features of the included papers is given, and finally, the main findings of the systematic review are summarised narratively. An overview of the included records can be found in Table 2.

3.1. Descriptive features of the included studies

The articles included in this review principally cover bilingual education in Europe (40) and the United States of America (USA) (31). Four address South America, two were conducted in Asia, and two include countries on different continents. Regarding the form of bilingual education, 42 articles specifically address CLIL, 25 deal with bilingual education in general and twelve consider CBI, dual-language education or immersion. As concerns study design, 32 were theoretical works, three were literature reviews and 44 were empirical contributions, 25 of which had a qualitative design, with interviews being the most used data source. Furthermore, four studies had a quantitative design through questionnaires/surveys. Last, 15 studies had a mixed-method design, with most including both interviews and questionnaires/surveys, sometimes in combination with additional data collection methods (Table 2).

3.2. Main findings

3.2.1. Teachers' competences in bilingual education

Our RQ addresses how BETs' competences are conceptualised within research. The systematic literature review covered 79 reports, which will be described in more detail. First, the competences defined in BETs' competence frameworks, profiles or overarching teacher training designs are examined in terms of overall convergence in the light of the theoretical background (Fig. 1). Second, further theoretical and empirical work on individual competences is presented. In the subsequent summary of the results for our RQ, we relate the review results to the theoretical background model and depict the resulting extended professional competence model for bilingual education teachers (Fig. 3).

The systematic literature review revealed a total of 24 competence frameworks or profiles. Of these 24 reports on competences, 17 describe teachers' competences for CLIL (e.g., Dafouz-Milne et al., 2010; Novotná et al., 2001; Zhorabekova, 2015) and two for dual-language settings (Guerrero & Lachance, 2018; McFarland, 1995). The remaining five cover overall BETs' competences in the USA (Blum Martinez & Baker, 2010; Calderón, 1997; Menken & Antunez, 2001), the Netherlands (European Platform, 2012) and Germany (Schauwienold-Rieger, 2012). Except for Novotná et al. (2001), who focus on teachers' competences for CLIL in the subject of mathematics, all papers have a cross-curricular focus, which supports the universal applicability of the resulting competence list. Table 3 illustrates the outcome of the comparison of the frameworks. In total, content analysis reveals 16 converging competences; only competences found in at least five reports were included. Not all frameworks mention the listed competences directly, but they can be deduced from the context or are emphasised as training needs. Within the table, these cases are marked with brackets around the check mark.

3.2.1.1. General competences for bilingual education teachers. To begin with, the eight top competences in Table 3 represent general sets

Table 2
List of included reports.

Author(s), Year	Bilingual Education Format	Country	Type of Article/ Study Design	Languages (Study or Program)	(Aspired) School Setting	Teachers' Subjects	Study Sample Background
Aiello et al. (2017)	CLIL	Italy	empirical study (mixed methods: language assessment, questionnaire, interviews, observations)	English & Italian	secondary schools	variety of subjects (mostly math, physics or humanities subjects)	subject teachers who will teach CLIL courses
Alfaro (2019)	dual-language education	USA	theoretical work	English & additional language	dual-language education, not specified more closely	not specified	X
Alfaro and Bartolomé (2017)	bilingual education	USA	theoretical work	English & Spanish	not specified	not specified	X
Aquino-Sterling (2016)	bilingual education	USA	empirical study (qualitative: transcripts of lesson descriptions)	English & Spanish	K-12 bilingual schools	not specified	students aiming to become BETs
Aquino-Sterling and Rodríguez-Valls (2016)	bilingual/ dual-language education	USA	theoretical work	English & Spanish	K-12 bilingual schools	not specified	X
Banegas et al. (2020)	CLIL	Latin America	literature review (2008–2018)	English & Latin America community's L1	primary, secondary and higher education	not specified	X
Barrios and Milla Lara (2020)	CLIL	Spain	empirical study (mixed methods: questionnaires, focus group interviews)	English & Spanish	primary and secondary schools	variety of subjects (mostly science or social science)	students, teachers and parents participating in CLIL programs
Bertaux et al., 2010	CLIL	Europe	theoretical work	not specified	not specified	not specified	X
Blum Martinez and Baker (2010)	bilingual education	USA	theoretical work	English & additional language	primary and secondary education	not specified	X
Briceño et al. (2018)	bilingual education	USA	empirical study (qualitative: interviews)	English & Spanish	not specified	not specified	bilingual teacher candidates (heritage Spanish speakers)
Calderón (1997)	bilingual education and English as a second language	USA	theoretical work	English & Spanish	primary and secondary education	not specified	X
Cammarata and Tedick (2012)	immersion education	USA	empirical study (qualitative: interviews, lived experience descriptions)	English & Spanish or French	primary, secondary and high school	math, history and social studies	immersion teachers
Cinganotto (2016)	CLIL	Italy	theoretical work	Italian & additional language	secondary schools	not specified	X
Cinganotto and Cuccurullo (2015)	CLIL	Italy	theoretical work	Italian & additional language	secondary schools	not specified	X
Coonan (2011)	CLIL	Europe	empirical study (qualitative: questionnaire, interviews)	not specified	not specified	not specified	non-language CLIL teachers
Cruz (2000)	immersion education	USA	empirical study (qualitative: interviews, journals, observations, program documents)	English & Spanish	middle school	social studies, math and/or reading, language arts or English	mainstream and bilingual teachers

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Table 2 (*continued*)

Author(s), Year	Bilingual Education Format	Country	Type of Article/ Study Design	Languages (Study or Program)	(Aspired) School Setting	Teachers' Subjects	Study Sample Background
Custodio-Espinar (2020)	CLIL	Spain	empirical study (quantitative: questionnaire)	English & Spanish	primary and secondary schools	second language variety of subjects	in-service CLIL teachers
Czura et al. (2009)	CLIL	Poland	empirical study (qualitative: classroom observations, interviews)	English & Polish	secondary schools	not specified	bilingual education coordinators, teachers and students
Dafouz-Milne et al. (2010)	CLIL	Europe	empirical study (qualitative: classroom observations)	Catalan, Czech, Dutch, English, French, German, Italian, Luxembourgish & Spanish	secondary and vocational education	not specified	CLIL content and language teachers, teacher trainers and academics
Dale et al. (2011)	CLIL	Netherlands	theoretical work	English & Dutch	secondary education	not specified, but examples given for various subjects	X
Durán-Martínez et al. (2016)	CLIL	Spain	empirical study (quantitative: questionnaire)	English & Spanish	primary and secondary schools	not specified	in-service non-language CLIL teachers (novices, experienced and experts)
Ekiaka-Oblazamengo (2018)	bilingual education	USA	empirical study (qualitative: interviews, syllabi)	English & Spanish or Hebrew	primary and secondary schools	not specified	BET educators and teacher candidates
Escobar Urmeneta (2010)	CLIL	Spain	theoretical work	English & Spanish & Catalan	secondary education	not specified	X
Escobar Urmeneta and Walsh (2017)	CLIL	Spain	empirical study (qualitative: in-class interactions from corpus)	English & Spanish & Catalan	secondary schools	science, biology and math	transcribed CLIL classroom interactions
European Platform (2012)	CLIL	Netherlands	theoretical work	English & Dutch	secondary bilingual schools	not specified	X
Fernández Costales and Lahuerta Martínez (2014)	CLIL	Spain	theoretical work	primarily English & additional language	primary, secondary and tertiary education	not specified	X
García Laborda and Alcalde Peñalver (2020)	CLIL	Spain	empirical study (mixed methods: questionnaire)	English & Spanish	not specified	not specified	prospective teachers of CLIL and non-CLIL courses
Giraldo and Murcia (2018)	bilingual education	Colombia	empirical study (mixed methods: survey, questionnaire, interviews, researcher journals)	English & Spanish	not specified	language education in general	professors, language teacher education expert and pre-service teachers
Gnutzmann (2015)	CLIL	Germany	theoretical work	foreign language & German	not specified	not specified	X
Grant (1997)	bilingual education	USA	empirical study (mixed methods: test scores, questionnaires)	English & Spanish	primary and secondary education	not specified	examinees of a Spanish proficiency exam, bilingual teacher experts
Guerrero (1997)	bilingual education	USA	theoretical work	English & Spanish	primary and secondary education	not specified	X

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Table 2 (*continued*)

Author(s), Year	Bilingual Education Format	Country	Type of Article/ Study Design	Languages (Study or Program)	(Aspired) School Setting	Teachers' Subjects	Study Sample Background
Guerrero (1998)	bilingual education	USA	theoretical work	English & Spanish	primary and secondary education	not specified	X
Guerrero (1999)	bilingual education	USA	theoretical work	English & Spanish	primary and secondary education	not specified	X
Guerrero and Lachance (2018)	dual-language education	USA	theoretical work	English & additional language	primary and secondary education	not specified	X
Hartono (2016)	bilingual education (immersion)	Indonesia	theoretical work	primarily English & Indonesian	kindergarten, primary and secondary education	not specified	X
Hernández & Alfaro (2020)	dual-language education	USA	theoretical work	English & Spanish	PreK-12 dual-language education	not specified	X
Hillyard (2011)	CLIL	Europe	theoretical work	English & additional language	primary, secondary, tertiary education	not specified	X
Hood (2020)	two-way immersion education	USA	empirical study (qualitative: interview, focus group discussions, journal prompts, pre-/postsurvey)	English & Spanish	primary and middle schools	not specified	pre-service teachers, practicing immersion teachers, and immersion administrators
Johannessen and Bustamante-López (2002)	bilingual education	USA	theoretical work[a]	English & Spanish	primary and secondary education	not specified	X
Knudson (1998)	bilingual education	USA	empirical study (mixed methods: survey, observations, interviews)	English	primary, middle and high schools	multiple subjects, single subjects, special education and bilingual education	student teachers and interns
Koopman et al. (2014)	CLIL	Netherlands	empirical study (qualitative: lesson observations, interviews, language teaching wall activity)	English & Dutch	secondary schools	biology, geography, history, maths and multicultural studies	experienced CLIL subject teachers
Leisen (2015)	CLIL	Germany	theoretical work	German as foreign language	not specified	not specified	X
Lemberger and Reyes-Carrasquillo (2011)	bilingual education and English as a second language	USA	empirical study (qualitative: questionnaire, interviews, classroom observations)	English & Spanish	kindergarten, primary and secondary schools, and adult education	variety of subjects (social studies, science, math or language arts, English-*as*-a-second-language)	bilingual education and English as a second language teachers
Limerick (2020)	intercultural bilingual education	Ecuador	empirical study (qualitative: participant observation, interviews, document and curriculum analysis)	Kichwa & Spanish	Indigenous schools, not specified more closely	not specified	examiners und examinees of the Kichwa exam

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Table 2 (*continued*)

Author(s), Year	Bilingual Education Format	Country	Type of Article/ Study Design	Languages (Study or Program)	(Aspired) School Setting	Teachers' Subjects	Study Sample Background
Liu and Rutledge (2020)	bilingual education	USA	empirical study (mixed methods: survey, diary entries)	English, but some participants also speak Spanish or an Indigenous language	K-5 bilingual classrooms but K-12 teachers	not specified	monolingual and bilingual pre-service teachers
Lopriore (2020)	CLIL	Italy	empirical study (qualitative: questionnaire, interviews, documentary analyses)	English & Italian	upper secondary schools	different subjects, not specified more closely	future CLIL subject teachers
Lorenzo (2008)	CLIL	Spain	empirical study (mixed methods: adapted texts)	English & Spanish	secondary school	bilingual history	second language teachers familiar with CLIL
Maljers et al. (2007)	CLIL	Europe	theoretical work	not specified	not specified	not specified	X
Maroney and Smith (2000)	bilingual education	USA	empirical study (qualitative: interviews)	English & Spanish	primary and secondary education	not specified	Mexican-trained BETs
Marsh et al. (2011)	CLIL	Europe	theoretical work	not specified	not specified	not specified	X
Mattheoudakis and Alexiou (2017)	CLIL	Greece	empirical study (qualitative: interviews)	English & Greek	primary and secondary schools	only specified for two teachers (math and physics)	English language teachers, a generalist teacher and subject teachers teaching CLIL
McFarland (1995)	bilingual education and English as a second language	USA	theoretical work	English	not specified	not specified	X
Méndez García and Pavón (2012)	CLIL	Spain	empirical study (qualitative: interviews)	Spanish & French	primary and secondary schools	geography, history, economy, music, physical education, science, physics and chemistry	language assistants, content teachers and language teachers
Menken and Antunez (2001)	bilingual education	USA	empirical study (mixed methods: survey, state-level licensure requirements, requirements of bilingual education training programs)	English & additional language	primary and secondary education	not specified	bilingual teacher education programs
Morton (2016)	content-based instruction	Spain	empirical study (qualitative: observations, interviews, video-recordings)	English & Spanish	secondary school	science (biology)	bilingual education subject teacher and students
Morton (2018)	CLIL	Spain	empirical study (qualitative: classroom interactions)	English & Spanish	secondary school	science (biology, chemistry)	bilingual education subject teachers and students
Murillo (2018)	dual-language education	USA	empirical study (qualitative: testimonios)	English & Spanish	primary schools (and middle school) with dual-	not specified	Latinx dual-language teachers

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Author(s), Year	Bilingual Education Format	Country	Type of Article/ Study Design	Languages (Study or Program)	(Aspired) School Setting	Teachers' Subjects	Study Sample Background
					language programs		
Novotná et al. (2001)	CLIL	Czech Republic	theoretical work[b]	English & Czech	primary and secondary education	math	X
Pavón and Ellison (2013)	CLIL	Europe	theoretical work	not specified	not specified	not specified	X
Pavón et al. (2020)	CLIL	Spain	empirical study (quantitative: questionnaire)	English & Spanish	primary and secondary schools	not specified	content teachers, language teachers and language assistants involved in CLIL
Pavón and Rubio (2010)	CLIL	Spain	theoretical work	Spanish & additional language	primary and secondary education	not specified	X
Pérez Agustín (2019)	CLIL	Europe/ Spain	literature review	English & Spanish	not specified	not specified	X
Pérez Cañado (2016a)	CLIL	Europe	empirical study (mixed-methods: questionnaires)	English, Spanish or additional language	primary and secondary schools, universities, provincial educational administration	not specified	pre- and in-service CLIL subject or language teachers, teacher trainers and coordinators
Pérez Cañado (2016b)	CLIL	Europe	empirical study (mixed-methods: questionnaires)	English, Spanish or additional language	primary and secondary schools	not specified	in-service CLIL teachers
Pérez Cañado (2018)	CLIL	Europe, Latin America, Asia	theoretical work	not specified	not specified	not specified	X
Pistorio (2009)	CLIL	Argentina	theoretical work	English & first language	not specified	not specified	X
Porcedda and González-Martínez (2020)	CLIL	international	systematic literature review	English or Spanish articles	primary, secondary and higher education	not specified	X
Quero Hermosilla and González-Gijón (2017)	bilingual education	Spain	empirical study (quantitative: questionnaire)	English & Spanish/ Castilian	secondary school	not specified	non-language subject teachers
Quezada and Alexandrowicz (2019)	dual-language education	USA	theoretical work	English & additional language	not specified	not specified	X
Relaño Pastor and Poveda (2020)	CLIL	Spain	empirical study (qualitative: classroom interactions, ethnographic data, interviews)	English & Spanish	secondary school	biology, religion and ethics	Language assistants, content teachers and coordinator
Robinson (2020)	CLIL	Italy	empirical study (mixed methods: questionnaires)	English & Italian	secondary education	variety of subjects, from science to philosophy	future CLIL subject teachers
Rodríguez-Valls et al. (2017)	dual-immersion education	USA	theoretical work	English & Spanish	primary and secondary education	social studies (history), Spanish language arts	X
Román et al. (2019)	bilingual education	USA	empirical study (mixed methods: survey)	English & Spanish	pre-kindergarten, kindergarten, primary and secondary schools	variety of subjects, not specified in detail	Spanish speaking bilingual teachers

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Table 2 (*continued*)

Author(s), Year	Bilingual Education Format	Country	Type of Article/ Study Design	Languages (Study or Program)	(Aspired) School Setting	Teachers' Subjects	Study Sample Background
Schauwienold-Rieger (2012)	bilingual education	Germany	empirical study (mixed methods: document analysis, expert surveys, interviews)	German & additional language	primary and secondary schools	not specified	kindergarten teachers, teachers, students, parents, principals, teacher educators, vocational teachers, business representatives
Szwed and González-Carriedo (2019)	bilingual education	USA	empirical study (qualitative: interviews)	English & Spanish	early childhood, primary and secondary education	not specified	preservice BETs
van Kampen et al. (2020)	CLIL	Netherlands	empirical study (qualitative: interviews)	English & Dutch	secondary schools and CLIL teacher education programs	variety of subjects (drama, biology, math, physical education, fine arts and social studies)	CLIL practitioners and specialists
Varghese (2004)	bilingual education	USA	empirical study (qualitative: field notes, interviews, documents)	English & Spanish	primary and secondary education	not specified	instructors, assistant and administrator of the professional development, apprentice bilingual teachers
Vomvoridi-Ivanović (2012)	bilingual education	USA	empirical study (qualitative: field notes, oral comments, observations)	English & Spanish	primary education (with primary and secondary education teachers)	math	bilingual Latin@ pre-service teachers
Zhorabekova (2015)	CLIL	Kazakhstan	empirical study (mixed methods: survey, observation, interviews, testing analysis, documents)	English, Kazakh & Russian	not specified	not specified	(pre-service) language and subject teachers

[a] Report includes three studies, however, the studies' results were not relevant for this literature review.
[b] Report also refers to an empirical study, however, the study's results were not relevant for this literature review.

of skills relevant to all teachers but with a particular importance in bilingual education. There is perfect or almost perfect agreement in all reports regarding competences referring to pedagogical/psychological knowledge (Baumert & Kunter, 2013), such as knowledge of learning processes, methodology, material design, assessment and classroom management (CM), highlighting their importance for bilingual education. Moreover, the review singled out content knowledge as a competence of crucial importance (20 reports). In comparison with these frequently listed competences, information and communication technology (ICT) and knowledge of research only appear in 14 and 13 reports, respectively, which indicates that these competences are deemed less relevant.

3.2.1.2. Specific competences for bilingual education teachers. The bottom eight competences displayed in Table 3 represent specific sets of teacher skills required for bilingual education. All reports identify language proficiency, as well as knowledge on how to merge content, language and learning, as essential for BETs. The latter falls into the domain of pedagogical content knowledge while language proficiency makes up a new knowledge domain within our theoretical background (resulting in a new category in Fig. 3). Most reports indicate that language proficiency requires more than Basic Interpersonal Communication Skills (Cummins, 1979) or standard language (Cummins, 2017) and should include Cummins' (1979; 2017) Cognitive Academic Language Proficiency or academic language (Bertaux et al., 2010; Dale et al., 2011; Hillyard, 2011; Marsh et al., 2011; Pavón & Ellison, 2013; Pérez Cañado, 2018) and

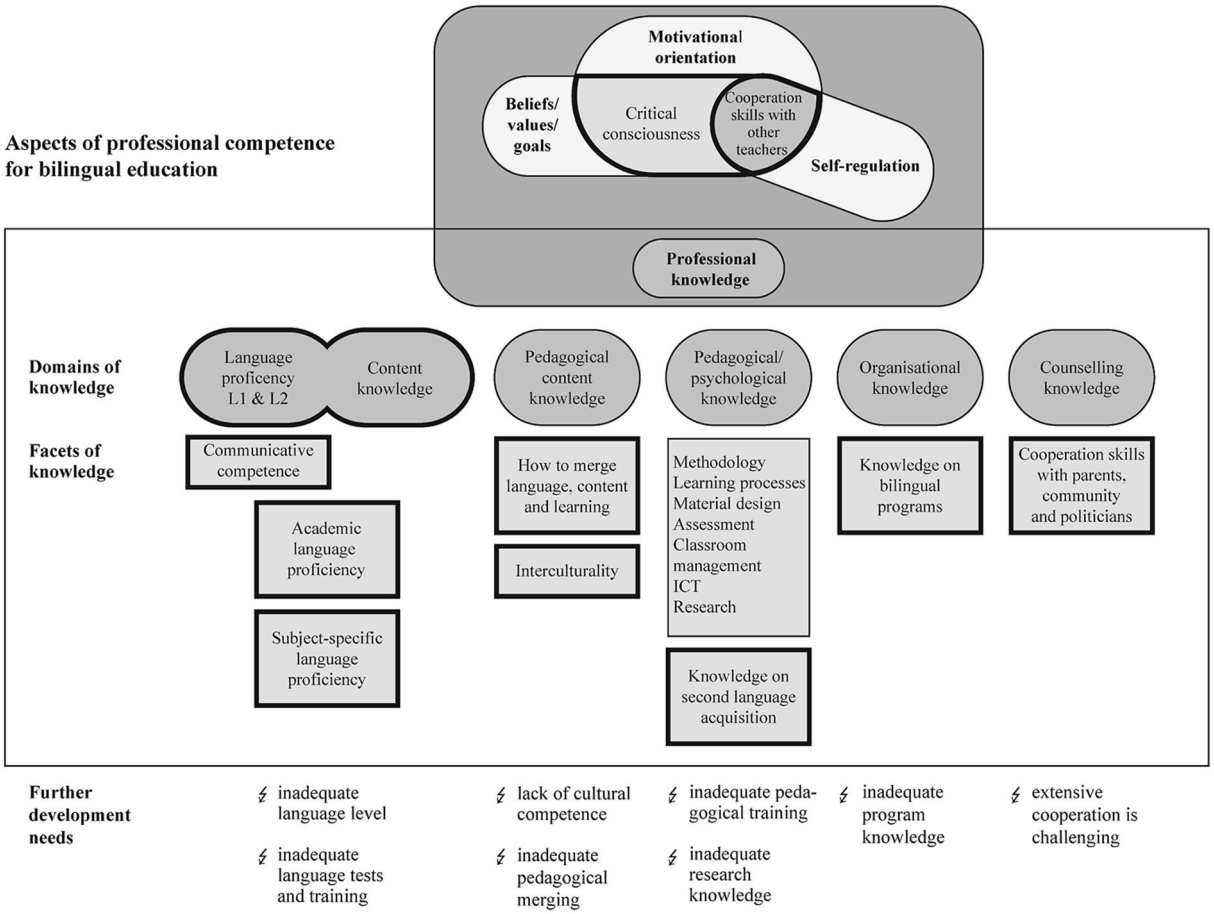


Fig. 3. Extended model of teachers' professional competence (Baumert & Kunter, 2013) for the context of bilingual education.

subject-specific language proficiency (Cinganotto, 2016; Gnutzmann, 2015). At the same time, only five reports (Blum Martinez & Baker, 2010; Calderón, 1997; Menken & Antunez, 2001; Novotná et al., 2001; Pistorio, 2009) underline the importance of language proficiency in the teacher's mother tongue, with the other authors only referring to second or foreign language proficiency. However, in this regard, Guerrero and Lachance (2018) point out that there is a research gap on the effect of teachers' bilingualism and biliteracy on the bilingual and biliteracy abilities of their students. Therefore, it is hardly surprising that most reports do not state a fixed level of required language proficiency but, instead, highlight the effective use of language within the bilingual classroom as a central precondition for successful bilingual teaching (e.g., Escobar Urmeneta, 2010; Guerrero & Lachance, 2018; Leisen, 2015; Maljers et al., 2007; Novotná et al., 2001). Nevertheless, knowledge related to (second) language (L2) acquisition, as part of pedagogical/psychological knowledge, is only mentioned in 17 reports, although this competence is intuitively necessary for effective language learning as it helps teachers to understand the processes to learn a language.

Concerning cooperation skills, 21 of 24 reports (see Table 3) include these as an expected competence for BETs. The majority focusing on working together with other teachers (Calderón, 1997; Cinganotto, 2016; Dafouz-Milne et al., 2010; Dale et al., 2011; Escobar Urmeneta, 2010; European Platform, 2012; Leisen, 2015; Lopriore, 2020; Pavón & Ellison, 2013; Pérez Cañado, 2018; Pistorio, 2009; Schauwienold-Rieger, 2012). This competence is not included in the theoretical background model and therefore added as a new aspect within our bilingual education competence model (Fig. 3). Other reports focus on cooperation with parents or the community (Blum Martinez & Baker, 2010; McFarland, 1995; Menken & Antunez, 2001) or with all stakeholders of bilingual education (Bertaux et al., 2010; Coonan, 2011; Guerrero & Lachance, 2018; Hillyard, 2011; Maljers et al., 2007; Marsh et al., 2011). These cooperation skills are classified as part of counselling knowledge in our competence model (Fig. 3).

Interculturality and critical consciousness as competences related to cultural aspects came up in 18 (e.g., Bertaux et al., 2010; Hillyard, 2011) and five reports (e.g., Calderón, 1997; Guerrero & Lachance, 2018; Menken & Antunez, 2001), respectively. Interculturality describes the promotion of cultural awareness and cultural exchange (Bertaux et al., 2010) and falls into the domain of pedagogical content knowledge. In comparison, critical consciousness encompasses awareness and reflection on (oppressive) language or cultural ideologies and potential advocacy for bilingual students' rights (Palmer et al., 2019) and only occurs in reports from the USA. Although critical consciousness is not integrated in the theoretical background model, it can be categorised as an intersection between beliefs/values/goals and motivational orientation (Fig. 3).

Table 3
Comparison of competence frameworks.

Competences Knowledge on …	Bertaux et al. (2010)	Blum Martinez and Baker (2010)	Calderón (1997)	Cinganotto (2016)	Coonan (2011)	Dafouz-Milne et al. (2010)	Dale et al. (2011)	Escoba Urmeneta (2010)
… methodology	✓	✓	✓	✓	✓	✓	✓	✓
… learning processes	✓	(✓)	✓	(✓)	✓	(✓)	✓	✓
… material design	✓	(✓)	(✓)	✓	✓	(✓)	✓	✓
… assessment	✓	✓	✓	✓	✓	✓	✓	✓
CM	✓	(✓)	✓	×	✓	✓	(✓)	✓
CK	(✓)	✓	✓	✓	(✓)	✓	(✓)	✓
… ICT	✓	×	✓	✓	×	×	(✓)	×
… research	✓	✓	✓	×	×	×	(✓)	×
Language proficiency	✓	✓	✓	✓	✓	(✓)	✓	✓
… how to merge content, language & learning	✓	✓	✓	✓	✓	✓	✓	✓
Cooperation skills	✓	✓	✓	✓	✓	✓	(✓)	✓
Interculturality	✓	✓	✓	×	✓	✓	×	×
L2 acquisition	✓	✓	✓	×	(✓)	×	✓	×
… bilingual programs (implementation)	✓	✓	✓	×	✓	×	×	×
Willingness to learn, motivation	✓	×	×	×	×	×	×	(✓)
Critical consciousness	×	✓	✓	×	×	×	×	×

Competences Knowledge on …	European Platform (2012)	Gnutzmann (2015)	Guerrero and Lachance (2018)	Hillyard (2011)	Leisen (2015)	Lopriore (2020)	Maljers et al. (2007)	Marsh et al. (2011)
… methodology	✓	✓	✓	✓	✓	✓	✓	✓
… learning processes	✓	(✓)	(✓)	✓	✓	✓	(✓)	✓
… material design	✓	✓	✓	✓	✓	✓	✓	✓
… assessment	✓	✓	✓	✓	✓	✓	×	✓
CM	✓	×	(✓)	✓	✓	(✓)	✓	✓
CK	(✓)	✓	×	×	✓	(✓)	×	✓
… ICT	✓	×	×	✓	×	✓	×	(✓)
… research	×	×	(✓)	(✓)	✓	✓	×	✓
Language proficiency	✓	✓	✓	✓	✓	✓	✓	✓
… how to merge content, language & learning	✓	✓	✓	✓	✓	✓	✓	✓
Cooperation skills	✓	×	✓	✓	✓	✓	✓	✓
Interculturality	✓	✓	✓	✓	×	(✓)	✓	✓
L2 acquisition	(✓)	×	✓	✓	✓	✓	×	✓
… bilingual programs (implementation)	✓	×	✓	✓	×	×	×	✓
Willingness to learn, motivation	×	×	×	✓	✓	(✓)	×	✓
Critical consciousness	×	×	✓	×	×	×	×	×

Competences Knowledge on …	McFarland (1995)	Menken and Antunez (2001)	Novotná et al. (2001)	Pavón and Ellison (2013)	Pérez Cañado (2018)	Pistorio (2009)	Schauwienold-Rieger (2012)	Zhorabekova (2015)	✓ & (✓) in sum
… methodology	✓	✓	✓	✓	✓	✓	✓	✓	24/24
… learning processes	✓	✓	✓	✓	(✓)	✓	✓	(✓)	24/24
… material design	(✓)	✓	(✓)	✓	✓	✓	(✓)	×	24/24
… assessment	✓	✓	✓	✓	✓	✓	✓	✓	23/24
CM	✓	×	(✓)	✓	✓	(✓)	(✓)	✓	21/24
CK	×	✓	✓	✓	✓	✓	✓	✓	20/24
… ICT	×	×	✓	(✓)	✓	✓	✓	✓	14/24
… research	✓	✓	×	✓	✓	×	×	×	13/24
Language proficiency	(✓)	✓	✓	✓	✓	✓	✓	✓	24/24
… how to merge content, language & learning	✓	✓	✓	✓	✓	✓	✓	✓	24/24
Cooperation skills	✓	✓	×	✓	✓	✓	✓	×	21/24
Interculturality	✓	✓	✓	✓	✓	×	✓	×	18/24
L2 acquisition	✓	✓	(✓)	✓	×	✓	✓	×	17/24
… bilingual programs (implementation)	✓	✓	×	✓	✓	(✓)	(✓)	✓	15/24
Willingness to learn, motivation	×	×	×	×	✓	×	✓	✓	10/24
Critical consciousness	✓	✓	×	×	×	×	×	×	5/24

CM (classroom management), CK (content knowledge), ICT (information & communication technology).
✓ mentioned (✓) indirectly mentioned × not mentioned.

Knowledge on bilingual programs and their implementation (as part of teachers' organisational knowledge), is mentioned in just over half the reports, whereas the presence of a fundamental willingness to learn or motivation (motivational orientation in the theoretical background model) is only listed in ten reports (Table 3), all of which address CLIL settings (e.g., Hillyard, 2011; Leisen, 2015; Lopriore, 2020; Zhorabekova, 2015) or German bilingual education (Schauwienold-Rieger, 2012).

3.2.1.3. Competence overview as basis for reflection. It is noteworthy that this list of 16 competences is only treated as an indication of BETs' competences and as a point of reference for self-reflection for development needs because the examined reports do not declare that teachers must have all of them. More precisely, Hillyard (2011) points out that acquiring professional competence in bilingual education requires practice and assimilation time. Development over time was also highlighted in a study by Durán-Martínez et al. (2016), who assessed the perceptions of ideal CLIL teacher competences of Spanish expert and novice in-service content CLIL teachers (n = 151). While experience increased the scepticism toward the quality of existing bilingual course material, it also increased appreciation of cooperation, innovation and bilingual education in general (Durán-Martínez et al., 2016). With this example of empirically inquired competences, the following section further examines the 16 competences for BETs by extending them through theoretical and empirical contributions on individual competences. Including this first example, a total of 59 reports (four of them were already included as framework or profile) are discussed in the following section.

3.2.2. Individual competences in theoretical and empirical research

3.2.2.1. Language proficiency. In addition to its great significance in the frameworks above, **language proficiency** is also reflected in the following 20 reports, often in combination with **(pedagogical) content knowledge**. First of all, Morton (2016) reinforces that language proficiency alone is insufficient for successful bilingual education and argues that a specialised pedagogical content knowledge is needed. Therefore, he introduces content and language knowledge for teaching (CLKT) as a conceptual heuristic for CLIL teachers, which belongs to the facet of how to merge language, content and learning (Fig. 3). CLKT encompasses common and specialised language knowledge, knowledge of language and students, and knowledge of language and teaching. Later, he elaborates in detail on the aspect of language proficiency for CLIL and proposes language knowledge for content teaching (LKCT) as an overarching concept (Morton, 2018). LKCT, in turn, consists of common language knowledge and specialised language knowledge for content teaching. The former describes linguistic competences shared with non-teaching professionals in the content area whereas the latter is teaching-specific knowledge on how to use language to convey the content in a way which makes it highly accessible to students (Morton, 2018). This request – for teaching-specific components of language competence – is supported by Escobar Urmeneta and Walsh (2017), who analyse Walsh's (2011) classroom interaction competence (CIC) in the context of CLIL extracts. In principle, CIC describes the ability to use interaction to mediate and assist learning. Their analysis yields the harmonisation of language use and pedagogical goals, the creation of learning spaces for students, and the shaping of students' language through feedback as important components of CIC in CLIL (Escobar Urmeneta & Walsh, 2017). Although Hartono (2016) uses the term 'communicative competence' instead of CIC, she also emphasises that, apart from general language proficiency, the ability to direct efficient discourse and meaning negotiations in class is crucial for BETs. A comparison between bilingual interns and student teachers in other fields showed that the former possess clearer theoretical orientations towards second language learning and teaching; thus, their orientation is functional and prioritises communicative competence (Knudson, 1998). These teaching-specific language competences illustrate a facet of BETs' language proficiency and are depicted as communicative competence within our extended competence model (Fig. 3).

In addition to the reference to cognitive academic language proficiency in some frameworks, Guerrero (1999) identifies academic language proficiency as important for BETs and defines it as a "speciali[s]ed register of language that teachers develop over time as they engage in educational activities" (p. 58). However, in his papers from the 1990s, he also sees obstacles that BETs encounter during the acquisition of this proficiency as the U.S. society and schools favour bilingual students' transition to the dominant majority language (Guerrero, 1997). This often makes them replace their native language with English (Guerrero, 1999). In addition to teachers' own schooling experiences, students' language preferences can make it difficult for pre-service teachers to use content-specific Spanish (Vomvoridi-Ivanović, 2012). Two effects of abiding U.S. bilingualism ideologies are, thus, bilingual pre-service teachers' lack of confidence (Briceño et al., 2018) and low self-efficacy in regard to their Spanish (academic) language skills (Szwed & González-Carriedo, 2019). Similarly, anxiety concerning language use and low self-perceived language proficiency frequently occurs among Italian CLIL content teachers (Aiello et al., 2017; Robinson, 2020). Overall, Hood (2020) points out that teachers, administrators and pre-service teachers recognise academic language proficiency as being of central importance in bilingual education and recommend study time abroad, which also supports cultural competence.

So far, this review has only considered the notion of language proficiency in classroom teaching. Aquino-Sterling (2016) expands this notion in his pedagogical Spanish competences (PSC) for bilingual education. The notion of PSC is very similar to Hartono's (2016) communicative competence because it highlights the language and literacy competences needed for effective teaching in the second language. Additionally, PSC include "meeting the professional language demands of working with students, colleagues, administrators, parents, and the larger bilingual school community" (Aquino-Sterling, 2016, p. 51). The significance of this additional language register is also elucidated within the overview of competence frameworks, most of which include cooperation skills. Therefore, Aquino-Sterling and Rodríguez-Valls (2016) stress that acquiring pedagogical and teaching-specific language competences is required for bilingual teacher education. Both academic and subject-specific language proficiency are depicted as facets of language proficiency and content knowledge within our competence model (Fig. 3).

While language proficiency itself is a challenging requirement for BETs, **language proficiency tests** in the USA are also criticised

in the literature. For example, Guerrero (1997, 1998) criticises bilingual teacher certification and claims it is of a low standard since its tests are designed for transitional education rather than for sustained native-language instruction. Grant's (1997) criticism goes further: she states that most states in the USA do not even have specialised language proficiency tests. Johannessen and Bustamante-López (2002) further claim that Spanish proficiency tests are not adequate for bilingual education purposes since they lack academic Spanish and do not include representative tasks for bilingual classrooms. Moreover, Lemberger and Reyes-Carrasquillo (2011) show that bilingual education and English as a second language teacher certification is challenging, but not necessarily a barrier to teaching. They also highlight the low validity of certification tests to signal teacher quality because in classroom observations teachers showed teaching qualities and practices which could not be measured in the tests.

Besides, a recent study shows that language standardisation in the writing part of the Indigenous Language Proficiency Exam in Ecuador did not account for existing language variations, which discriminates against speakers of such variations (Limerick, 2020). This shortcoming is especially important as minority language preservation is one of the goals of bilingual education named by Wright and Baker (2017), however, "tensions exist relating to the ultimate goals of bilingual education, even within the bilingual education community" (Varghese, 2004, p. 234).

All in all, the quantity of reports on language proficiency emphasises its central importance for bilingual education. Therefore, bilingual teacher education is demanded to ensure access to academic language registers and a sensitisation toward teachers' own positions within society (Blum Martinez & Baker, 2010). Beyond that, Szwed and González-Carriedo (2019) point out that bilingual teacher education is obliged to counteract English language hegemony, to reaffirm language varieties and to foster the development of both languages, for example through translanguaging. Translanguaging can be defined as "the deployment of a speaker's full linguistic repertoire without regard for watchful adherence to the socially and politically defined boundaries of named (and usually national and state) languages" (Otheguy et al., 2015, p. 283). These recommendations found in the literature are supported by Vomvoridi-Ivanović (2012), who highlights the importance of access to academic Spanish for prospective teachers but emphasises that the most pressing concern is that prospective teachers become used to practising Spanish in their subject. Last, Robinson (2020) and Aiello et al. (2017) elaborate that the development of language proficiency within a training program also needs to address psychological aspects to reduce language anxiety.

3.2.2.2. Pedagogical content knowledge. In addition to the abovementioned language specific-extensions to **pedagogical content knowledge**, Banegas et al. (2020) stress that professional development on how to merge content, language and learning for bilingual education needs to take the teaching-learning context into account. For instance, Lorenzo (2008) looks at CLIL teachers' approaches to adapting a foreign-language text about history. Teachers' approaches differed greatly; some simplified it, elaborated or created a new discourse entirely. Consequently, teachers need to be educated in how to modify content for language learners and about the advantages of the different approaches.

3.2.2.3. Critical consciousness. A third individual competence which is intensively debated in the reviewed reports is **critical consciousness**, which involves overcoming deficit language ideologies (Freire, 2021) and consists of both ideological reflection regarding language and culture and advocacy for students and bilingual education (Palmer et al., 2019). Even though critical consciousness was the least mentioned of the 16 competences included in all the competence frameworks (Table 3), the number of additional reports addressing it indicates that it is required for BETs. As a start, language ideologies are dealt with in three additional reports. First, in a study analysing BETs' attitudes toward varieties of Spanish commonly spoken by students, language purism ideologies and discrimination against users of such varieties were found (Román et al., 2019). Teachers' perspectives on (language) minority students and resulting discrimination were also investigated by Maroney and Smith (2000), who show that teachers who received their education in Mexico and train to become BETs in the USA still have deficit beliefs, for example arguing that minority students' culture gives reason to their failure in school. Furthermore, research showed that pre-service teachers underestimate the influence of accents and gender-specific language learning differences and overestimate job opportunities for bilinguals (Liu & Rutledge, 2020).

In contrast, Murillo (2018) found that negative experiences with language ideology in the USA can also motivate individuals to become BETs to support bilingual or immigrant students; therefore, they implicitly act as social change agents. In comparison, European CLIL teachers emphasise the challenge of, and an interest in, the bilingual teaching approach as their driving force (Lopriore, 2020; Mattheoudakis & Alexiou, 2017).

In his theoretical paper, Alfaro (2019) recommends actively developing critical consciousness during teacher education, for instance through the inclusion of a cultural wealth model, which values different cultural backgrounds and challenges deficit perspectives (Alfaro & Bartolomé, 2017). Other researchers propose the integration of social justice education (Ekiaka-Oblazamengo, 2018) or an assessment of teachers' cultural proficiency (Quezada & Alexandrowicz, 2019). The latter is defined as "an individual's, or a group's, belief system that holds students' cultural backgrounds of language, race, gender, socioeconomic status as assets on which one is to construct their educational experiences" (Quezada & Alexandrowicz, 2019, p. 186). Concerning advocacy, Varghese (2004) found that teacher educators expect all BETs to be advocates; however, she stresses that the acquisition of advocacy is a complex process which necessitates a focus on local and contextual learning within professional development.

3.2.2.4. Cooperation skills. **Cooperation skills between teachers** provide an additional aspect of BETs' professional competence (Fig. 3) with four reports. Relaño Pastor and Poveda (2020) investigate the cooperation between BETs and native language assistants and reveal problems with the establishment of a professional order between the parties based on language. In comparison, Méndez García and Pavón (2012) find teacher cooperation between content teachers and language assistants to be motivating and encouraging

for teachers. At the same time, they identify a lack of training on effective teacher cooperation because the collaborations considered were based on intuition alone. Cruz's (2000) study of the professional network in one middle school highlights that cooperation between mainstream and BETs can be beneficial for teaching practice in general. However, for this approach, teachers require extra time to plan together, which needs to be incorporated into the operating structure of the school (Cruz, 2000). Additionally, Rodríguez-Valls et al. (2017) propose horizontal cooperation between content teachers and language teachers because one teacher cannot fulfil all the roles and competences alone, namely (academic) language proficiency, content knowledge and teaching expertise. Last, Durán-Martínez et al. (2016) see the mentoring of novice teachers by experienced BETs as a possibility for bilingual teacher education to support cooperation and teaching quality.

3.2.2.5. Pedagogical/psychological knowledge. The following three sections describe six reports on BETs' **pedagogical/psychological knowledge**. The first two studies focus on teachers' CLIL pedagogy. In van Kampen et al.'s (2020) study, teachers and CLIL specialists agree that student-centred teaching, scaffolding, opportunities for interaction and a mixture of writing and speaking tasks are central for CLIL. Comparably, Koopman et al. (2014) look at CLIL teachers' language pedagogical knowledge and find that the classroom activities teachers are able to deploy consist mainly of word-level activities, with a wider range of language learning left out. Their findings also show a lack of knowledge about language acquisition as well as monotonous correction methods. Participants in van Kampen et al.'s study (2020) also considered language feedback and assessment of language to be the most challenging aspects of CLIL. Two further studies addressing language assessment present contradictory results. García Laborda and Alcalde Peñalver (2020) found that most participating CLIL pre-service teachers do not regard CLIL assessment as more difficult than general foreign language education assessments. In comparison, Giraldo and Murcia (2018) find high demands for training in the design of assessment instruments and enormous interest in methodologies and assessment in bilingual education.

Custodio-Espinar (2020) examines CLIL lesson methodological planning competence through the self-assessment of 383 primary and secondary CLIL teachers. In this study, all secondary teachers, except for advanced language teachers, show a need for planning competence improvement as well as training in formative and summative evaluation (Custodio-Espinar, 2020). This systematic literature review will not go deeper into the topic of empirical research on effective bilingual education methods, as doing so is beyond its scope and, indeed, could fill a whole review on its own. Moreover, listing successful methodological approaches could lead to the wrong conclusion, namely that teaching is an objectifiable craft that only requires "the mastery of technical components [although it really is] a highly contextuali[s]ed process" (Gay & Kirkland, 2003, p. 182). Furthermore, Hurajova (2019) indicates that various factors influence students' language competence and that methodology plays an important but not decisive role. Nevertheless, during the screening of the almost 4300 titles and abstracts, a strikingly high number of mentions of scaffolding, translanguaging, code-switching and the implementation of digital tools and media indicates the overall importance of methodological considerations in the field of bilingual education.

The last individual competence in the domain of pedagogical/psychological knowledge is digital competence. Cinganotto and Cuccurullo (2015) argue that CLIL teachers' professional profile must include digital competence, or, rather, technological pedagogical content knowledge (Mishra & Koehler, 2006), as the basis for the successful implementation of digital media, specifically videos, in the CLIL classroom.

3.2.2.6. Training needs. To conclude the findings, we next present the inferences of 14 papers on BETs' overall training needs and implications for bilingual teacher education, which are depicted as further development needs in the extended competence model (Fig. 3). First, regarding **BETs' training needs,** Pavón et al. (2020) found a general demand of balanced methodological and linguistic training. Furthermore, they discovered a need for CLIL training for content teachers, due to a lack of confidence in their CLIL teaching, and that language and content teachers should be trained jointly with language assistants to foster teamwork. Additionally, Pavón et al. (2020) emphasise the importance of continuous professional development, due to the dynamic and innovative character of CLIL. Quero Hermosilla and González-Gijón's (2017) findings show a lack of methodological training and general openness for language training. However, participating teachers voiced reluctance to take up training offers if they had to bear the costs themselves. Similarly, inadequate CLIL methodology training is the weakness most frequently reported in another study (Barrios & Milla Lara, 2020). Last, Fernández Costales and Lahuerta Martínez (2014) consider the theory-practice gap of methodological training at universities the main problem of teacher training.

The training needs of pre-service and in-service teachers, teacher educators and coordinators on a European scale are highlighted by Pérez Cañado (2016a, 2016b). These results reveal that there are training needs in all the examined areas: linguistic and cultural competence, knowledge of CLIL theory (research, programs, implementation, policies), methodology, material and continuous professional development. Overall, the linguistic and cultural competence training needs were less severe, whereas the need for theoretical knowledge of CLIL and continuous professional development was remarkable (Pérez Cañado, 2016a; 2016b). The results of another study also show teachers' uncertainties and, therefore, training needs regarding CLIL theory and methodology (Pavón & Rubio, 2010). Furthermore, Pavón and Rubio (2010) name the implementation of the integrated curriculum as a critical training field. In comparison, practising teachers emphasise training needs in the areas of language, methodology, materials and assessment, for instance, work experiences abroad, cooperation with language teachers or foreign language assessment (Coonan, 2011). Moreover, organisational, interpersonal and cooperation skills are also highlighted as a challenge for CLIL teacher training (Pérez Cañado, 2018). For CLIL teachers in Poland, Czura et al. (2009) identified CLIL-specific training needs with respect to subject-specific training, professional exchanges, and a demand for professional development in cooperation skills and international exchange programs. A Europe-wide literature review of CLIL teachers' training needs (Pérez Agustín, 2019) reveals that teachers' language proficiency,

methodological skills and knowledge of material design and assessment, as well as their cooperation skills, are capable of improvement. A further systematic review by Porcedda and González-Martínez (2020) on lacks in European CLIL training shows that prior CLIL training, pedagogical/educational training and instructional/planning processes are mentioned most frequently. The reviewed articles recommended training in CLIL, the inclusion of ICT and cooperative practices as measures to counter these lacks.

In comparison, training needs in the USA range from language proficiency, because of restricting language ideologies, to an overall need for bilingual training and preparation programs due to a current lack of bilingual faculty (Hernández & Alfaro, 2020). Cammarata and Tedick (2012) point out the need to increase immersion teachers' knowledge on how to integrate content and language in light of the misconception that teaching content in another language automatically leads to language learning. Therefore, training on language acquisition processes and more opportunities for collaboration are required (Cammarata & Tedick, 2012).

3.2.3. Summary of findings

All in all, the comparison of 24 competence frameworks or competence overviews resulted in a list of 16 competences for BETs with an overall good convergence. Additional theoretical or empirical research focused, in particular, on language proficiency, critical consciousness, pedagogical/psychological knowledge and training needs and yielded some implications for bilingual teacher education which are elaborated on below. Based on the model by Baumert and Kunter (2013), Fig. 3 summarises the review results though the depiction of an extended bilingual education professional competence model. The added and highlighted aspects (in bold) illustrate bilingual education-specific competences as we described them above. To begin with, critical consciousness and cooperation skills with other teachers are depicted as intersectional aspects of professional competence. In addition, deepened language proficiency and content knowledge which include teaching- and subject-specific facets are needed for bilingual education, regardless of teachers' educational background, i.e. whether they are language or content teachers. Other facets like knowledge on how to merge language, content and learning as well as interculturality fall into the category of bilingual education-specific pedagogical content knowledge. Moreover, except for knowledge on second language acquisition which is (at least for subject-oriented teachers) specific for bilingual education, all aspects of pedagogical/psychological knowledge are already included in the Baumert and Kunter (2013) model for general teaching. However, due to the emphasised lack thereof in the field of bilingual education, extensive pedagogical knowledge seems necessary for bilingual teachers. Furthermore, knowledge on bilingual programs is depicted as facet of organisational knowledge. Last but not least, cooperation skills with parents, the community or politicians are presented as additional facet of counselling knowledge.

4. Discussion

This systematic literature review closely analysed 79 individual reports to examine how the competences of BETs in secondary schools are conceptualised within research. The discussion which follows recapitulates the results of teachers' competences and connects them with other findings in different bilingual teacher education contexts.

The analysis revealed an overall convergence of international competence frameworks or training programs and resulted in a list of **16 competences** which can be considered essential for worldwide bilingual education. Many of these competences were already present within our underlying theoretical background model of teachers' competences (Baumert & Kunter, 2013); however, BETs require even more and high-level competences. The 16 competences can most likely be generalised to bilingual education since most of the included competence frameworks or overviews were either not directed at a target group (e.g., secondary education) or were created explicitly for both secondary and primary education. Furthermore, all frameworks have a cross-curricular focus (except for that of Novotná et al. (2001), who focus on the subject of mathematics), which can also be seen as an indication of the transferability and validity of the competences found.

The excess of **CLIL competence frameworks** within our timeframe of the last 25 years could be due to the fact that CLIL was implemented relatively late – in the 1990s – in Europe (compared to the USA, for instance); hence, empirical CLIL research began to increase from mid-2000 (Dalton-Puffer et al., 2014). Furthermore, in the USA, many teacher preparation programs and, therefore, competence frameworks focus on transitional bilingual education (Guerrero & Lachance, 2018), which were excluded from this review due to their implementation in primary education and the fact that they do not value the long-term preservation and development of bilingual capabilities.

We now discuss individual BETs' competences in more detail. The literature review showed that **critical consciousness** as a competence is, to date, only considered in the bilingual curricula of the USA. The amount of research regarding this concept has been increasing lately, in addition to its mentions in the frameworks. In the literature, possible means to foster critical consciousness were, for example, international exchanges (Alfaro & Quezada, 2010; Anthony-Stevens & Griño, 2018), personal exchanges (Fitts & Weisman, 2010; Hernández, 2017) or art- or drama-based pedagogies (Caldas, 2019). Other research showed the importance to leave space for teachers' development of language advocacy, especially in areas where there are still restrictive language policies, in order to preserve young people's language rights and bilingualism (Dubetz & Jong, 2011; Ramírez et al., 2016). Similarly, language awareness has started to receive increasing attention in Europe as well. For example, Sierens et al. (2018) conclude that language awareness interventions can help students develop "positive attitudes towards linguistic diversity and more favourable perceptions of diverse languages and of speakers of these languages" (p. 78). In qualitative research, BETs' own histories and their social background were documented to be connected to their language ideologies (Varghese & Snyder, 2018), which, in turn, are likely to influence instructional practices (Briceño, 2018). Accordingly, the situatedness of language and language ideologies, and the unfolding of and actively dealing with one's implicit ideologies and practices, can help to develop a professional identity (Zúñiga, 2019) and offers "an opportunity to break the cycle of oppression and [promote] the development of teachers as critical pedagogues" (Ostorga & Farruggio,

2020, p. 1234). These findings show that critical consciousness is increasingly discussed and also debated in the literature.

Although some competence frameworks, due to their length, have been criticised as extensive wish lists which can overstrain teacher education curricula (Morton, 2016), the comprehensive framework overview and the competence model (Fig. 3) of this systematic review can help teachers to reflect on their competences and increase their interest in further professional development. One might wonder why **reflection** is not added as a competence in this framework overview even though it is explicitly listed in some frameworks (e.g., Pérez Cañado, 2018; Escobar Urmeneta, 2010). There are at least two reasons: first, reflection is already partly included in the competence of lifelong learning and motivation. Second, it can be argued that reflection is an inherent part of teacher competences anyway (e.g., Beauchamp, 2015; Hatton & Smith, 1995), and therefore does not need to be part of a specific bilingual education competence framework. Nevertheless, it should be emphasised that reflection on personal development is necessary, in particular, because, for instance, not all countries in Europe require special teacher qualifications for CLIL (Eurydice, 2008). Furthermore, teachers need to continuously develop their competences when teaching in bilingual education and their "expertise is progressive by nature" (Jäppinen, 2006, p. 24). The first encounter with bilingual education might negatively affect teaching and cause fear and insecurities, in particular regarding linguistic competences, but these effects could be mitigated over time (Moate, 2011). Professional development and reflection can also help to further improve transition and assimilation because "the key to any future vision for bilingual education is teacher development" (Pérez Cañado & Ráez Padilla, 2015, p. 7).

Overall, our literature review illustrates the importance of BETs' **language proficiency** but also that such proficiency is frequently inadequate. Since teachers must have more advanced linguistic capabilities than learners in order to have two-fold language competence, that is, the ability to use language to explain content appropriately and to be aware of language features at the same time (Guerrero & Lachance, 2018), the emphasis on linguistic competences is reasonable. Moreover, it is recommended for teachers preparing for bilingual education to develop an understanding and awareness of academic (Cummins, 2017) or school language (May & Wright, 2007). Actively fostering students' language learning is sometimes even disregarded, as not all teachers are aware of the additional focus on language in CLIL (Hüttner et al., 2013). Hence, Jong and Barko-Alva (2015) highlight that teachers need to understand their changing role as instructors who integrate content and language. This in turn, requires an understanding of subject-specific literacies and their assessment (Whitehead, 2007). However, research shows that regular courses of education can result in BETs being poorly prepared to teach in the medium of, for instance, academic Spanish (Guerrero, 2003). Accordingly, this recommendation to foster teachers' awareness of academic language in bilingual education is also supported by Sutterby et al. (2005), who stress the need for pre-service BETs to experience academic Spanish in university to become comfortable with it.

In addition to more individual competences, that is, critical consciousness and language proficiency, our systematic review demonstrates the importance of teachers' **cooperation skills** for bilingual education (e.g., Marsh et al., 2011; Rodríguez-Valls et al., 2017). This result is confirmed in other bilingual teacher education contexts, which report on the successful testing of innovative approaches to foster cooperation skills. Examples include a bilingual mentor program between experienced English students and freshman primary education teachers (Arco-Tirado et al., 2018) and mentorship programs with bilingual pre-service teachers and experienced bilingual or ESL teachers (Riley et al., 2017) or with experienced former teachers and Indigenous teachers in rural areas (La Garza, 2016). Another example features cooperation between universities and schools during teacher education aimed at the improvement of teaching the natural sciences in a foreign language (Izquierdo et al., 2016). However, research on the experiences of bilingual and mainstream primary teachers shows that cooperation in schools can be difficult (Amos, 2020) and that bilingual (education) teachers are often not sufficiently appreciated (Valenta, 2009). Consequently, fostering cooperation in teacher education in general and establishing favourable cooperation at an early stage seem advantageous ways to support high-quality bilingual education.

Knowledge of research of bilingual education was underlined as essential in just over half the frameworks (e.g., Blum Martinez & Baker, 2010; McFarland, 1995; Pérez Cañado, 2018). Also, it was absent in the additional theoretical contributions and empirical studies that we found in our review. Once, knowledge of research was mentioned as a training lack (Pérez Cañado, 2016a; 2016b). Therefore, teachers' knowledge of bilingual education research can be seen as an important avenue for future training, research and practice. This recommendation that BETs should be aware of bilingual education research is also in line with research on primary BETs (Franco-Fuenmayor et al., 2015).

In conclusion, linguistic competences, critical consciousness, cooperation skills and knowledge of bilingual education research can be seen as important but also expandable competences for BETs. They have been shown to be a relevant part of professional education and reflection. Accordingly, these competences are also highlighted in the bilingual education professional competence model and can be seen as the central implications of this review which provide a starting point to improve bilingual teacher education programs.

From a **methodological point of view**, the systematic review revealed that most studies (37 out of 44 empirical studies) were based on interviews or questionnaires/surveys with self-assessment. Consequently, their explanatory power might be disputable. This is a major limitation of the studies included in the review and of the research field of bilingual education and therefore suggests an important avenue for future research. Further research implications are raised below in light of the review's limitations.

5. Limitations and conclusion

Despite its important contributions, the present systematic review has some limitations. Although the authors thoroughly examined different literature databases and aimed to reduce publication bias through a broad range of included publication criteria, relevant but (as yet) unpublished studies might have been left out. Second, while our broad timeframe allowed for a comprehensive overview of the field, it also raised some issues; for example, some examinations of language proficiency tests (e.g., Guerrero, 1997; Grant, 1997) might already be outdated. Thus, results with regard to language proficiency tests should be interpreted with caution. Moreover, except for these possibly outdated investigations, most reports do not include concrete recommendations for how to assess bilingual teachers'

competences. This is noteworthy since the competence model should go hand in hand with competence assessment. Third, merging the competence frameworks through content analysis left an interpretative margin for the authors since some frameworks used competence keywords without explanations (e.g., Dafouz-Milne et al., 2010; Lopriore, 2020). Fourth, although the authors aimed to only include reports on secondary bilingual education (frameworks, teachers or training programs) or the combination thereof with other education levels, some reports aimed at primary education might still be included because their actual target group was not always apparent within the text. Fifth, most of the included articles originated in the USA or Europe. Consequently, our results might be biased toward the specific conditions of these countries. The chosen languages (English and German) and the search string, which might not have been optimal since it elided terms like "CBI", "dual-language education" or "indigenous language education" (May, 2013), could have skewed our findings towards these areas and caused other relevant research from different cultural backgrounds to be left out. For a future systematic review, it is recommended to expand the search string by including the terms "CLIL teacher", "teacher program" as well as "CBI", "dual-language education", "two-way immersion" and "indigenous language education".

Nevertheless, (to our knowledge) this is the first systematic research which comprehensively investigates BETs' competence frameworks. Overall, the frameworks converge satisfactorily although some competences are still debatable, for example, critical consciousness is missing in Europe while investigated in the USA. With regard to BETs necessary competences and training needs, research over the last 25 years paid special attention to language proficiency, pedagogy, methodology (in particular) and cooperation skills.

The absence of quantitative studies measuring competence development without falling back on self-assessment can thereby be taken as an important implication for future research on bilingual education. Currently, the extent to which BETs fulfil the demands raised in the competence framework remains a rather open question. In consequence, the effectiveness of bilingual teacher education programs cannot yet be assessed. Nevertheless, as bilingual education, in particular dual-language education, "is widely supported by research as a highly effective […] approach that is associated with significant academic and linguistic benefits as well as amplified sociocultural and socioemotional competenc[es], or what might be called 21st century skills" (Guerrero & Lachance, 2018, p. 5), teachers need to be well equipped to effectively meet its challenges. Accordingly, future research on BETs' competences and competence development is needed.

On the whole, this systematic review contributes to the literature in at least two ways. First, it goes beyond existing literature reviews on European CLIL training lacks (Pérez Agustín, 2019; Porcedda & González-Martínez, 2020) by including research from several research contexts. Second, it closely investigates the different competence frameworks and systematically shows their convergence, resulting in an extended bilingual education professional competence model. Since most of the underlying reports do not include recommendations concerning competence assessments for bilingual teachers, developing an assessment strategy to complement the competence model would be a promising research area for the future. Nevertheless, the presented model can provide guidance or serve as sources of inspiration for bilingual educators around the world who want to improve their training approach. In conclusion, linguistic and skills training through personal and international exchanges, the inclusion of academic language and research on bilingual education are promising approaches for bilingual teacher education and should be promoted worldwide.

Additional information

The systematic literature review was preregistered on the platform Open Science Framework (https://doi.org/10.17605/OSF.IO/Z8JWU) using the Rmd-template by Schneider et al. (2022). The following adjustments were made to the original criteria used in the preregistration. The setting was originally primary, secondary and teacher education; however, primary education was excluded after all abstracts and titles had been screened, due to a high focus on transitional bilingual programs in the USA which were shown to be the least successful form of bilingual education (Collier & Thomas, 2004; Herrera, 2020). Nevertheless, articles including primary and secondary BETs were included because they are often trained together in the USA. Additionally, teacher-educators were removed from the specific target group since they were indirectly included through the setting of teacher education programs. Furthermore, the sub-questions of RQ1 were removed due to their intersecting characteristics. Last, due to the qualitative nature of most of the literature found, the originally-intended examination of the influence of bilingual teacher education programs on professional competence (RQ2) was discarded.

CRediT author statement

Luisa Scherzinger: Conceptualisation, Methodology, Investigation, Writing - Original Draft **Taiga Brahm**: Conceptualisation, Methodology, Supervision, Writing - Reviewing and Editing.

Declaration of competing interest

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Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.edurev.2023.100531.

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